15.1 A 3.9mW 200words/min Neural Signal Processor in Speech Decoding for Brain-Machine Interface

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Brain-machine interfaces (BMIs) are a promising technology that can be applied to AR/VR interfaces, neural prostheses, and machine control. Figure 15.1.1 shows BMI systems based on the source of decoded neural activities: visual stimulation, handwriting, and speech [1-3]. A visual-stimulation-based BMI [1] decodes intended characters by observing flickering targets, but an external stimulus is necessary. A handwriting-based BMI [2] converts attempted handwriting movements to characters, but the communication rate is low for natural speech. A speech-based BMI [3] translates speech attempts into words and this enables communication at a higher rate. In a speech-based BMI, a neural network (NN) infers the probability of each phone (the smallest unit of speech sound) being spoken according to the extracted brainwave features. The most likely sequence of words can be decoded based on the phone probability with a language model by employing beam search. Dedicated processors have been demonstrated for visual-stimulation-based and handwriting-based systems [4-5]. However, efficient hardware mapping for speech-based BMIs, which achieves the highest communication rate, has not been extensively explored due to the excessively high NN computational complexity. According to our experiment, it takes 5.3 seconds to decode one word on a high-end CPU with 65W for a speech-based BMI, which is infeasible for implanted devices. This work presents an energy-efficient neural signal processor for real-time speech decoding (with up to 200 words/min for conversational speech [6]).

Figure 15.1.2 shows the speech-based BMI workflow with added speech attempt detection and algorithm-architecture co-optimizations. In this work, a speech attempt is detected by analyzing neural signals from a massive array of channels. The key neural features, spike-band power (SBP) and threshold crossing rate (TCR), are then extracted in a frame of 80ms. A recurrent neural network (RNN) is usually used to process input features and to produce phone probability. In this work, a skim RNN [7], which is composed of a big RNN and a small RNN, is adopted to skip computations conditionally. Both the big and small RNNs are composed of five layers of gated-recurrent units (GRUs), but with different hidden dimensions. The Viterbi beam search algorithm [8] is applied to find the most likely sentence. The system is designed by optimizations across layers of abstraction. As selected in [3], the number of channels for speech decoding is 128. Of the 128 channels, 16 channels with the highest firing rates are selected for speech attempt detection. The model size of the RNN is minimized through sparse encoding and mixed-precision arithmetic. Non-zero (NZ) weights are encoded in value and index pairs to reduce the memory storage for sparse data. Most of the weights located around the peaks (inliers) with a narrower data range can be quantized to 4b and the remaining weights (outliers) are encoded in an 8b format. Compared with a neural network without sparse encoding in 8b, the overall memory storage is reduced by 80%.

Figure 15.1.3 shows the system architecture of the proposed neural signal processor, which includes a feature extractor, a neural network engine, a beam search engine, and a speech attempt detector. In the feature extractor, the neural signals are filtered by multiple bandpass filter banks. Then, the SBP and TCR are extracted as features for the succeeding RNN. The neural network engine is composed of an array of processing elements (PEs) for performing multiply-accumulate (MAC) operations. Activation functions (Sigmoid and Softsign) are implemented by a nonlinear unit. A skim unit is designed to evaluate the importance of input words. Based on the evaluation result, the hidden states are partially or fully updated by passing through a small RNN or a big RNN, respectively. In the beam search engine, the states are updated based on the states of the previous frame and those with lower likelihood scores are pruned for Viterbi decoding. For the speech attempt detector, the neural signals from the selected channels are identified to determine the speech attempt in an always-on mode. The other hardware modules are only activated when an attempt is detected, reducing the overall power dissipation by 46% through clock gating.

Figure 15.1.4 shows the details for the neural network engine. There is a trade-off between the computational complexity and the phone error rate. The design is optimized to minimize the computational complexity while maintaining the phone error rate by choosing a proper threshold. This reduces the number of operations by 38% with a phone error rate of 15.6%. In this work, partial sum (Psum) caching and computation reordering are proposed to reduce the processing latency for RNN. For partial sum caching, the computations for updating hidden states can be reduced by reusing the partial sum calculated in the previous frame. Since half of the data in hidden states remain the same, the pre-cached partial sum can be reused to reduce the computations in the next frame or the next layer. Compared to the implementation without Psum caching, the number of operations is reduced by 25%. For computation reordering, the consecutive zeros resulting from the ReLU6 activation function can be leveraged to reduce the latency by speculating zero output. The output of the GRU is zero if both partial sums (Psum1 and Psum2) in the candidate hidden state ($\hat{h}$) are non-positive and the hidden state of the previous frame is zero. If the output is determined as zero, all the remaining computations can be skipped by computation reordering, in which Psum1 and Psum2 are speculatively computed. Compared to the baseline order, the computation reordering technique can reduce the latency of the neural network by 55%.

Figure 15.1.5 shows the details of the processing element in the neural network engine and the beam search engine. The processing element is composed of one matching unit and 16 MAC units. For the matching unit, the input is first fetched based on the NZ index of the weight to form a pair of NZ weight and input, and then the pair is skipped if the input is zero. For the speech decoding dataset in [3] (with a sparsity of 62%), the processing latency can be reduced by 95% by applying sparse encoding to the weight (with a sparsity of 87.5%). For the MAC unit, a mixed-precision multiplier is deployed to support both 4b×8b and 8b×8b multiplications in an area-efficient way. It takes only one cycle to perform a 4b×8b multiplication in most cases and two cycles to perform an 8b×8b multiplication in very few cases. Compared to the MAC array with a full precision, the proposed mixed-precision design has 27% smaller area at a cost of less than a 3% increase in the average latency. In the beam search engine, the less likely states can be discarded by applying state pruning with an insertion sort engine [9]. However, the complexity of such a design grows linearly with the beam size, introducing high hardware complexity for a large-vocabulary language model (with up to 125,000 words). In this work, approximate sorting is proposed to reduce the hardware complexity by sorting the data over multiple bins, while the data within each bin remain unsorted. The uneven distribution of the likelihood score is addressed by allocating data to empty bins with pointers. Compared with the baseline design, the proposed architecture includes 16× fewer comparators. A state filter is used to retain the state with the maximum likelihood score among the states with the same previous words. The hardware complexity can be minimized by utilizing an XOR-based hash table. Compared with the direct-mapped realization, the area of the proposed state filter is reduced by 45%.

Figure 15.1.6 shows the verification results and performance comparison. Fabricated in 40nm CMOS, the chip integrates 3.2M logic gates in the core area of 2.22mm^2. The functionality of the chip is verified using the 56-hour dataset with 10,850 sentences and 4,219 words in [3]. The attempted speech of the subject is decoded into a sequence of phones (including a silent phone) and the words in the sentence can be identified by beam search. The phone [word] error rate is calculated by comparing the decoded sequence of phones [words] with those in the benchmark. This work achieves a phone error rate (PER) of 16.6% and a word error rate (WER) of 23.5%. This chip can be operated at 2 to 200MHz and the latency of decoding one word is 300ms (at 6.7MHz). Compared with state-of-the-art BMI processors [4-5], this work can support decoding for the maximum communication rate of 200 words/min, which is 16.7 to 42.6× faster than prior works [4-5], with power consumption of 3.9mW. This work is able to decode up to 125,000 words (in which 4,219 words are validated), a capability not achievable by prior works that can only decode up to 31 characters. The chip delivers energy efficiency of 0.9 words/mJ at 6.7MHz, 0.6V, which is comparable with prior works. Figure 15.1.7 shows the chip micrograph and chip summary.

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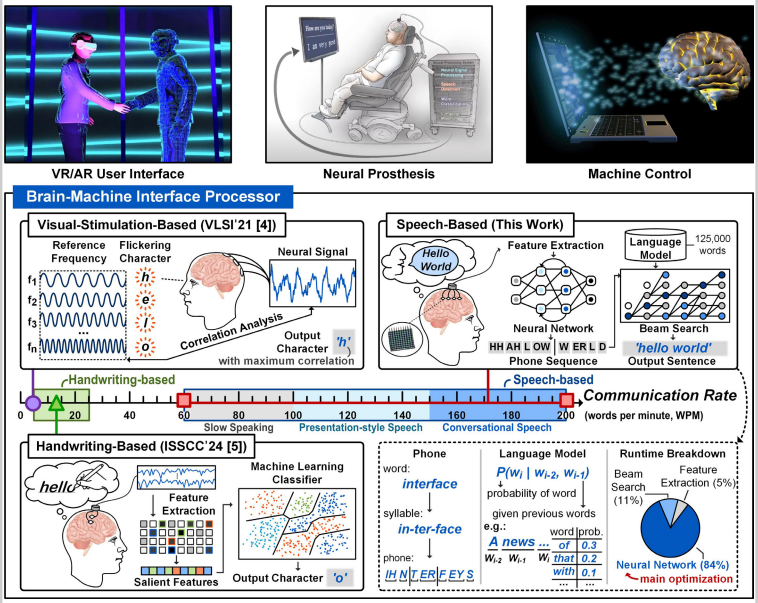


Figure 15.1.1: Brain-machine interface applications and BMI systems.

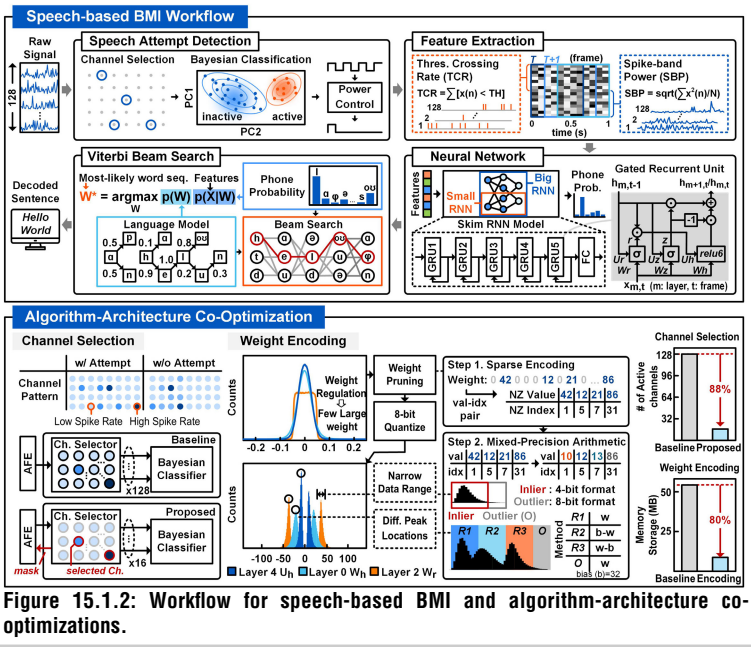


Figure 15.1.2: Workflow for speech-based BMI and algorithm-architecture co-optimizations.

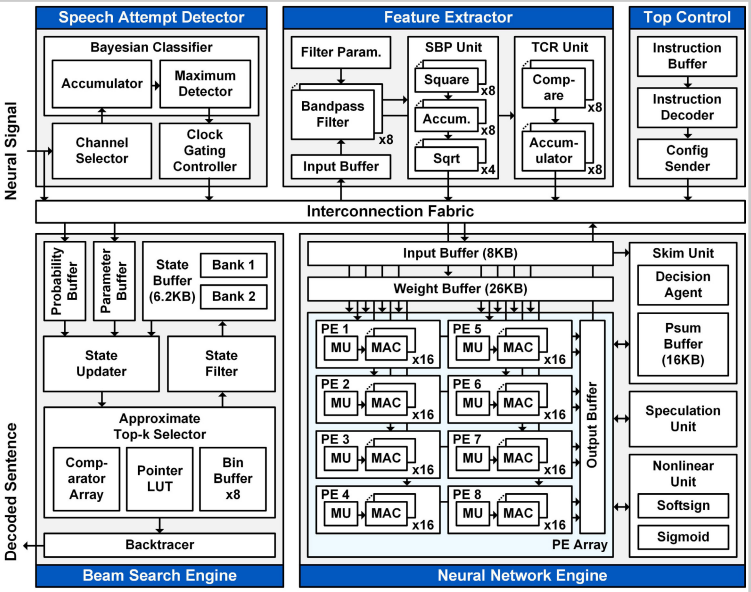


Figure 15.1.3: System architecture of the proposed neural signal processor for speech decoding.

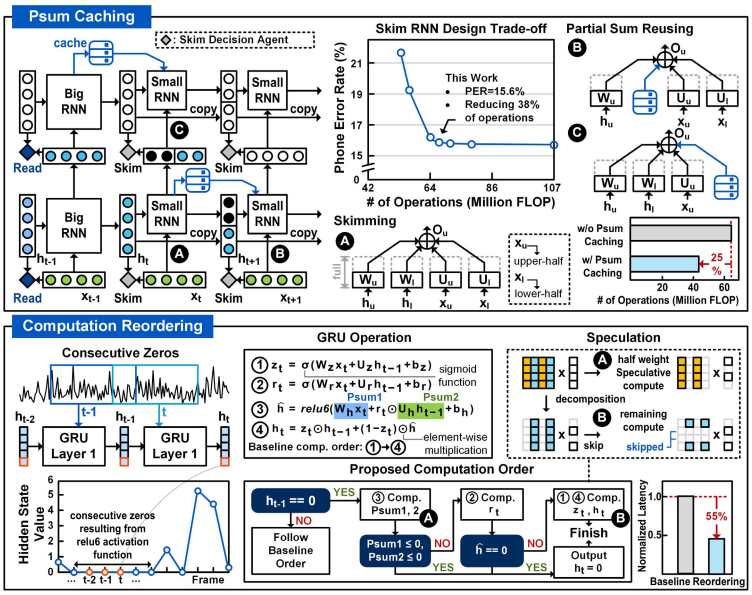


Figure 15.1.4: Architecture optimizations on the neural network engine.

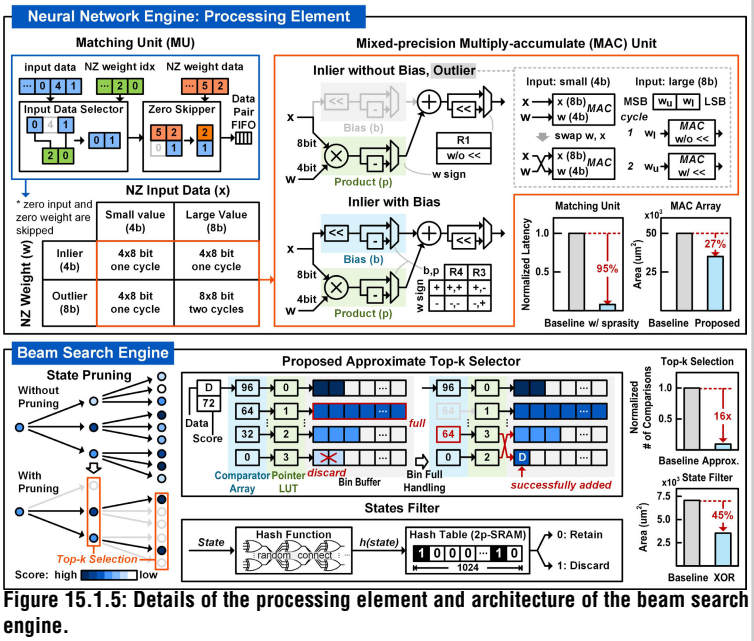


Figure 15.1.5: Details of the processing element and architecture of the beam search engine.

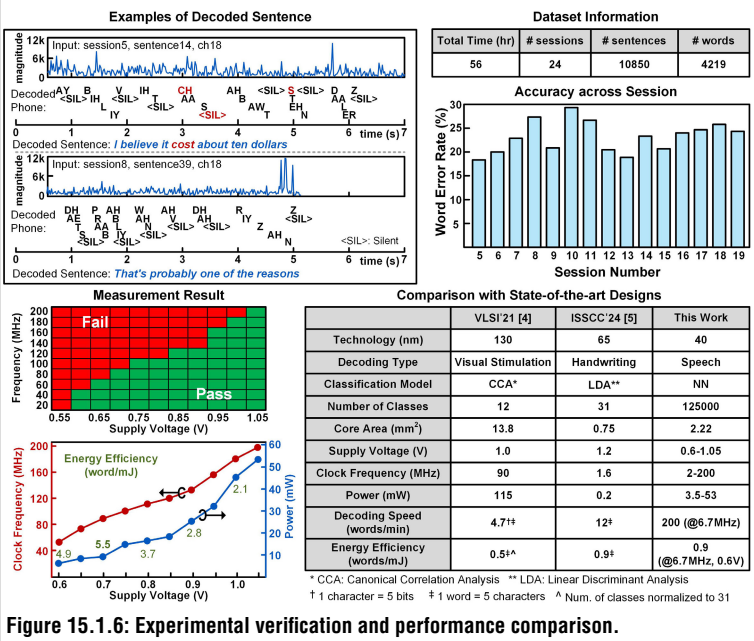
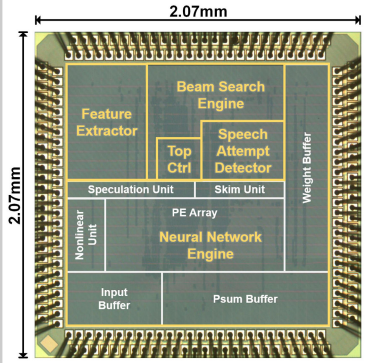


Figure 15.1.6: Experimental verification and performance comparison.



Technology		40nm 1P9M
Area	Die	2.07 x 2.07 mm^2
	Core	1.49 x 1.49 mm^2
Gate Count		3.2M
On-chip SRAM (KB)		76
Core VDD (V)		0.6-1.05
Frequence (MHz)		2-200
Power (mW)		3.5-53
Latency (ms/word)		9.5-950
Error Rate (%)		PER: 16.6 WER: 23.5
Decoding Speed* (words/min)		60-200

* limited by the feasible communication rate

Off-chip memory: 11MB for RNN, 290MB for language model

Figure 15.1.7: Chip micrograph and summary.

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